

HeatWatch AI

AI-Based Climate Intelligence System for Heatwave Monitoring, Prediction, and Early Warning

Historical Data Report

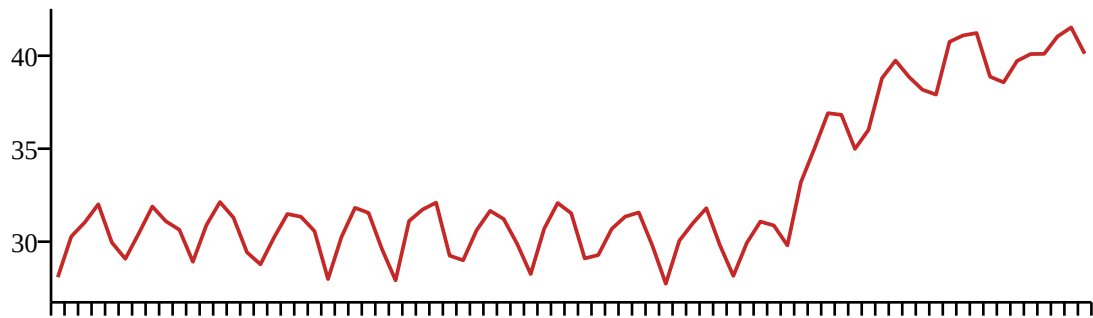
Location	Mumbai, Maharashtra
Coordinates	19.0760, 72.8777
Reporting period	2026-09-05 to 2026-09-19
Generated at (UTC)	2026-09-19 09:44:07
Display timezone	Asia/Kolkata

1. Weather Statistics

Metric	Value	Metric	Value
Observations stored	77	Avg temperature	32.72 °C
Max temperature	41.52 °C	Min temperature	27.74 °C
Avg humidity	69.25 %	Avg wind speed	13.71 km/h
Heat index (avg)	42.67 °C	Heat index (max)	65.9 °C

Heat index is reported only where the NWS calculation is applicable (77 of 77 observations). Data modes present: DEMO.

Temperature trend (stored observations)



2. Heatwave Risk Analysis

Predictions	17
Low risk	0
Moderate risk	11
High risk	6
Average severity	72.53 / 100
Peak severity	100.0 / 100
Data modes	DEMO

Risk assessments combine a weather-based severity model with a documented rule-based sentiment fusion layer. Model methodology and limitations are listed in the HeatWatch AI model card ([docs/model/model_card.md](#)).

3. Sentiment Analysis Summary

Posts analysed	180
Positive / Neutral / Negative	68 / 51 / 61

Heat-related posts	64
Heat-related distress signals	40
Average sentiment score	-0.03
Data modes	DEMO

General sentiment is distinct from heat-related distress; the distress count only includes heat-relevant posts with health/impact vocabulary.

4. Alert Summary

Generated at (UTC)	Level	Severity	Status
2026-09-16 10:00	HIGH	100/100	acknowledged
2026-09-17 10:00	HIGH	100/100	active
2026-09-18 10:00	HIGH	97/100	active
2026-09-19 09:39	HIGH	90/100	active

Total alerts in period: 4 (high: 4, moderate: 0). All alerts are application-generated warnings, not official advisories.

5. Data Source and Methodology Notes

- Weather observations: provider(s) DEMO; units: temperature in degC, wind in km/h, pressure in hPa.
- Heat index: NWS Rothfusz regression; reported only where applicability conditions hold.
- Sentiment model: VADER (pretrained, English social-media lexicon), compound score in [-1, 1].
- Severity model: weather-based scikit-learn RandomForest trained on the documented heat-index banding dataset, combined with a rule-based sentiment fusion layer. See the model card for evaluation metrics and limitations.
- Provenance: records marked DEMO are seeded demonstration data and do not represent real observations or real public posts.
- Report generated by HeatWatch AI; content is derived exclusively from the database records matching the selected filters.